Transformation to Microservices

Domain-Driven Design (DDD) Refactoring & Acceptance Criteria Solutions

1. Domain Context Mapping & Primary Entities

Patient Care Context

- **Patient Entity:** Demographic & contact info
- **Appointment Entity:** Schedule & booking state
- **Medical Record Entity:** Patient clinical history
- **Practitioner Entity:** Doctor/Nurse profiles

Billing & Info Context

- **Invoice Entity:** Itemized charges & balance
- **Insurance Claim Entity:** Provider claims
- **Payment Entity:** Receipts & transactions
- **Coverage Entity:** Policy eligibility data

Diagnostics Context

- **Lab Order Entity:** Ordered diagnostic tests
- **Test Result Entity:** Laboratory findings
- **Specimen Entity:** Sample tracking ID
- **Lab Technician Entity:** Assigned staff

2. User Stories for "Patient Care" Context

User Story 1: Self-Service Appointment Rescheduling

As a registered patient,
I want to reschedule my upcoming appointment online,
So that I can adjust my clinical visits to my schedule without calling reception.

Acceptance Criteria: Requests submitted within 24 hours trigger a late-change flag and notify the practitioner automatically.

User Story 2: Patient Medical History Retrieval

As an attending physician,
I want to view a patient's historical appointment summaries and diagnoses,
So that I can make informed clinical decisions during patient consultations.

Acceptance Criteria: Records load in under 2 seconds and restricted records require multi-factor authorization.

3. Acceptance Criteria in Gherkin Syntax

```
Scenario: Patient reschedules an appointment within 24 hours of window
  Given a patient has a scheduled appointment for "2026-10-15T10:00:00Z"
  When the patient requests a reschedule to "2026-10-16T14:00:00Z" less than 24 hours
before the original time
  Then the system should apply a late-change flag
  And emit an "AppointmentRescheduled" event to the Notification Service
  And display a confirmation message with updated details to the patient
```